MIL-B-50957(MU)
27 September 1971
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

BATTERY, THERMAL: M9

1. SCOPE

1.1 This specification covers one type of thermal battery, designated Battery, Thermal: M9, which is used to furnish electrical power to the Pershing warhead section for arming functions and for air burst fuzing.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-60902 - Soldering Electrical Connections for Nuclear Weapons Systems, Procedure for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ARMY BALLISTIC MISSILE AGENCY

10617679 - Thermal Battery, M9.

FSC 6135

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ORDNANCE CORPS

8856849 - Drum, Metal, Shipping, for Battery, Thermal: M9.

PICATINNY ARSENAL

8808362 - Packing and Marking for Storage and Shipments.

PUBLICATIONS

PICATINNY ARSENAL

EL-10617679 - Inspection Equipment List.
QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Materials and components. - Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence. See 6.2.) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly. - The thermal battery shall comply with all requirements specified on drawing (dwg.) 10617679, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article testing. - Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2.1).

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3.4 Initiator resistance. - The initiator resistance of the thermal battery shall be as specified in Table I. In addition, the resistance values specified in Table I shall be recorded (see 3.15). Testing shall be as specified in 4.4.1.

TABLE I

From Pin	To Pin	Resistance (ohms)	Test Identification Number (see 3.15)*
B	G	1.1 - 1.5	030001
C	F	1.1 - 1.5	030002

*Code numbers shall be recorded only for tests conducted at room temperature and not during environmental testing.

3.5 Indicator fuse resistance.

3.5.1 Before battery activation. - The indicator fuse resistance shall be as specified in Table II. In addition, the resistance value specified in Table II shall be recorded (see 3.15). Testing shall be as specified in 4.4.2.1.

TABLE II

From Pin	To Pin	Resistance (ohms)	Test Identification Number (see 3.15)*
A	E	1.0 max	041001

*Code numbers shall be recorded only for tests conducted at room temperature and not during environmental testing.

3.5.2 After battery activation. - The indicator fuse circuit resistance, measured between terminals A and E, shall be not less than 100,000 ohms after the battery cools to room temperature. Testing shall be as specified in 4.4.2.2.

3.6 Insulation resistance. - The insulation resistance of the thermal battery shall be not less than 20 megohms between the points listed in Table III. The insulation resistance shall be measured after a 10-second application of the test voltage. There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.4.3.

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TABLE III

From (+)	To (-)
AE	Case and all other pins
BG	"
CF	"
MJ	"

3.7 Leakage. - The thermal battery shall withstand the leakage test specified in 4.4.4 without evidence of leakage, distortion, or other damage.

3.8 Polarity. - The thermal battery shall be tested for correct polarity as specified in 4.4.5. Terminals M-J shall be positive and terminals K-L shall be negative.

3.9 No-fire current. - The battery initiators shall not fire when a direct current of 100 -0, +5 milliamperes (ma) is allowed to flow through each of the two initiators. The current shall flow through each initiator for a minimum period of one minute. Testing shall be as specified in 4.4.6.

3.10 Thermal battery operational tests. - During these tests, and unless otherwise specified, a constant load shall be applied across the battery terminals, and it shall consist of 13 ± 0.7 ohm with a 0.15 ohm resistance ± 5 per cent in series with each battery terminal and the 13 ohm load resistor. Battery voltage shall be measured across the 13 ohm load resistor using the method specified in 4.4.7.

3.10.1 Activation current. - The battery shall activate (as evidenced by a rise in the output voltage curve) when a direct current of 500 +0, -25 ma is allowed to flow through either one of the two initiators. Testing shall be as specified in 4.4.7.1.

3.10.2 Activation time. - The battery voltage shall be a minimum of 26 volts within a 5-second period after battery activation. Testing shall be as specified in 4.4.7.2.

3.10.3 Load test. - The battery shall function satisfactorily for a minimum period of 180 seconds after battery activation. During this load period, the battery voltage shall be not greater than 33 volts. After activation, the pulse loads, specified in (a) through (e) below, shall be shunted across the constant 13 ohm load resistor. After each pulse load, the battery voltage shall return to a minimum of 26 volts. In addition, the voltage values specified in Table IV (corresponding to the times of the pulse loads specified in (a) through (e) below) shall be recorded during applicable tests (see 3.15).

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During this load test, the recorded voltage curve shall be fairly smooth with no abrupt voltage changes (except during pulse loads). After this test, the battery shall comply with the requirements of 3.5.2. Testing shall be as specified in 4.4.7.3:

(a) At 5 +1, -0 second from activation, a load of 0.78 ohm +5, -0 per cent shall be applied for 50 +20, -0 millisecond (msec). The pulse load voltage shall be not less than 11 volts.

(b) At 40 +0, -1 second, a load of 0.46 ohm +5, -0 per cent shall be applied for 50 +20, -0 msec. The pulse load voltage shall be not less than 11 volts.

(c) At 41 +1, -0 second, a load of 2.5 ohms ± 5 per cent shall be applied for 50 +20, -0 msec. The pulse load voltage shall be not less than 15 volts.

(d) At 43 +1, -0 second, a load of 1.15 ohms +5, -0 per cent shall be applied for 1 +0.5, -0 second. the pulse load voltage shall be not less than 15 volts.

(e) At 120 +0, -1 second, a load of 2.5 ohms ± 5 per cent shall be applied for 50 +20, -0 msec. The pulse load voltage shall be not less than 20 volts. After the 50 msec pulse load duration, the pulse load and the 13 ohm constant load shall be removed from the battery. The open-circuit voltage shall be 28 +5, -2 volts for a minimum period of 60 seconds.

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Table IV

Time After Activation (seconds)	Battery Voltage	Test Identification Numbers (see 3.15) Room Temp.	High Temp.	Low Temp.	Temp. Shock	Vib-ration
5	11 min.	103001	110001	120001	130001	140001
40	11 "	103002	110002	120002	130002	140002
41	15 "	103003	110003	120003	130003	140003
43	15 "	103004	110004	120004	130004	140004
120	20 "	103005	110005	120005	130005	140005
180	28 +5 -2	103006 (A)	110006 (A)	120006 (B)	130006 (A)	140006 (B)

(A) Data recorded only during first article testing.

(B) Data recorded during first article and quality conformance tests.

3.11 High temperature. - The thermal battery shall withstand a high temperature of 160° ± 5°F., for a minimum period of 6 hours, without evidence of cracking, swelling, distortion, or other damage that may impair its intended operation. While at the high temperature (at the end of the high-temperature exposure period), the battery shall comply with the requirements of 3.4, 3.5.1, 3.6, 3.8, and 3.9. The battery to be activated shall comply with the requirements of 3.7, at room temperature, and of 3.10 after the battery has been returned to the high temperature. Testing shall be as specified in 4.4.8.

3.12 Low temperature. - The thermal battery shall withstand a low temperature of -65° ± 5°F., for a minimum period of 6 hours, without evidence of cracking, distortion, or other damage that may impair its intended operation. While at the low temperature (at the end of the low-temperature exposure period), the battery shall comply with the requirements of 3.4, 3.5.1, 3.6, and 3.9. The battery to be activated shall be warmed to room temperature and shall comply with the requirements of 3.8 and 3.7. It shall then comply with the requirements of 3.10 after the battery has been returned to the low temperature. Testing shall be as specified in 4.4.9.

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3.13 Temperature shock. - The thermal battery shall withstand alternate exposures to temperatures of 160° ± 5°F. and -65° ± 5°F., for a minimum period of 4 hours at each exposure, and there shall be no evidence of cracking, swelling, distortion, or other damage that may impair its intended operation. While at room temperature, the battery shall comply with the requirements of 3.4, 3.5.1, 3.6, 3.7, 3.8, and 3.9. The battery to be activated shall comply with the requirements of 3.10. Testing shall be as specified in 4.4.10.

3.14 Vibration. - The thermal battery shall withstand vibrations as specified in Table V without evidence of deterioration or damage that may impair its intended operation. After vibration testing, the battery shall comply with the requirements of 3.4, 3.5.1, 3.6, 3.7, 3.8, 3.9, and 3.10. Testing shall be as specified in 4.4.11.

TABLE V

Frequency Range cps	Double Displacement or Peak g Level ± 10% Tol.	Minimum Time Period to Cover Freq. Range (minutes)
10-30	0.20 inch	1.75
30-1000	8 g	2.0
1000-2000	10 g	3.75

3.15 Serialization. - A six-letter serial code shall be assigned to each serialized production item. The contractor shall obtain these serial codes by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SMUPA-NR2. The letter of request shall include: (a) exact nomenclature of the item as specified on the applicable drawing, (b) contract number or procurement request order number, and (c) number of serial codes required (unused codes must be returned). In addition to the normal inspection records, the contractor shall provide data feedback (see 6.1.2) in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the test identification numbers, are specified in the applicable paragraphs in this section.

3.16 Soldering. - All soldering shall be in accordance with MIL-S-60902.

3.17 Workmanship. - All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and

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sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Submission. - When the contract requires first article inspection, the contractor shall submit 10 consecutively produced thermal batteries to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. - The first article inspection samples shall be inspected for the defects specified in 4.3.2.1, using inspection methods contained therein, and for all requirements of Section 3 (also see Table VI) using the test methods specified in 4.4. The thermal batteries shall be tested in the order specified in Table VI. If any thermal battery fails to comply with the applicable requirements, the first article samples shall be rejected. Prior to submitting the samples to the testing facility designated (see 4.2.1), the contractor may perform any nondestructive test listed in Table VI (tests (a) through (f)).

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TABLE VI

Test	Requirement paragraph	No. of units tested	No. of units to be activated after each test*
(a) Initiator resistance	3.4	10	0
(b) Indicator fuse resistance (before battery activation)	3.5.1	10	0
(c) Insulation resistance	3.6	10	0
(d) Leakage	3.7	10	0
(e) Polarity	3.8	10	0
(f) No-fire current	3.9	10	0
(g) Thermal battery operational tests	3.10	2	2
(h) High temperature	3.11	8	2
(i) Low temperature	3.12	6	2
(j) Temperature shock	3.13	4	2
(k) Vibration	3.14	2	2

*Units to be activated shall comply with the requirements of the thermal battery operational tests (see 3.10).

4.2.3 Inspection equipment. - The inspection and testing specified in 4.2.2 shall be accomplished with the use of the approved gages and test equipment specified in 4.3.4.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. - A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. - A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table VII). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 per cent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

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TABLE VII

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	32	31-45	28
46-60	37	46-60	31
61-75	40	61-75	34
76-95	43	76-95	36
96-130	46	96-130	38
131-180	50	131-180	40
181-250	52	181-250	42
251-400	54	251-400	43
401 and over	56	401 and over	44

4.3.2.1 Classification of defects. - Critical, major, and minor defects are defined in MIL-STD-105.

4.3.2.1.1 Thermal Battery, M9 (see dwg. 10617679).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of mounting hole (4 holes)	Gage
102.	Flatness of bottom of mounting bracket with mounting plane (4 brackets)	Gage
103.	Perpendicularity of mounting hole with mounting plane (4 holes)	Gage
104.	Height between bottom of case and mounting plane, min	Gage
105.	Ability of connector to engage and disengage with mating receptacle	Manual/Gage
106.	Weight	Approved Weighing Device
107.	Height of pins in connector (see 6.3)	Visual
108.	Any connector pin bent or damaged	Visual
109.	Connector insert material cut or damaged	Visual
110.	External battery surface damaged	Visual

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Minor:

201. Height of mounting bracket, max. (4 brackets).....Gage
202. Protective finish missing or improper.....Visual
203. Identification plate missing, improperly marked, or improperly located.....Visual
204. Evidence of poor workmanship (see 3.17)...Visual

4.3.3 Inspection Testing. - The contractor shall perform all of the tests specified in this paragraph. The tests shall be performed in the following sequence:

	Test	Test Classification
(a)	Initiator resistance	Major
(b)	Indicator fuse resistance (before battery activation)	Major
(c)	Insulation resistance	Major
(d)	Leakage	Major
(e)	Polarity	Major
(f)	No-fire current	Major
(g)	Low temperature	Major
(h)	Vibration	Major

Of these tests, (g) and (h) are considered destructive and samples so tested shall not be returned to the lot.

4.3.3.1 Initiator resistance (see 3.4), indicator fuze resistance (before battery activation) (see 3.5.1), insulation resistance (see 3.6), leakage (see 3.7), polarity (see 3.8), and no-fire current (see 3.9). - Each thermal battery shall be subjected to these tests (see applicable paragraph in 4.4). If any battery fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.3.3.2 Low temperature (see 3.12) and vibration (see 3.14). - Two thermal batteries shall be selected from every lot of 25 batteries produced, or portion thereof. One of the batteries shall be subjected to the low-temperature test, and the other battery shall be subjected to the vibration test (see applicable paragraph in 4.4). If any battery fails to comply with the applicable requirements, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

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4.3.4.1 Critical and major defects and tests. - Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions as specified on Equipment List, EL-10617679. When either of the following conditions exists, the contractor shall design required equipment or specify suitable commerical equipment: (a) If any deviation from equipment or procedures, as specified on the EL, is desired (b) the code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2). The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions prior to use on the contract (see 6.4).

4.3.4.2 Minor defects and tests. - Acceptance inspection equipment for minor defects is not specified on ELs. The acceptance inspection equipment, operating instructions, and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods and procedures. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature ($77^{o} \pm 10^{o}F.$). Caution shall be observed while performing all tests (see 6.5).

4.4.1 Initiator resistance. - The resistance of each battery initiator shall be measured with a Government-approved tester. During these measurements, the initiator current shall be limited to a maximum of 10 ma, and the initiator resistance shall comply with the requirements of 3.4.

4.4.2 Indicator fuse resistance.

4.4.2.1 Before battery activation. - The resistance of the indicator fuse shall be measured with a Government-approved tester. During these measurements, the indicator fuse current shall be limited to a maximum of 10 ma, and the indicator fuse resistance shall comply with the requirements of 3.5.1.

4.4.2.2 After battery activation. - The open-circuit resistance of the indicator fuse circuit shall be measured with a Government-approved tester, and it shall comply with the requirements of 3.5.2.

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4.4.3 Insulation resistance. - The insulation resistance of the thermal battery shall be measured by using a Government-approved insulation resistance tester at an applied voltage of 500 ± 50 volts direct current with polarity as indicated in Table III. The battery shall comply with the requirements of 3.6 during this test.

4.4.4 Leakage. - The thermal battery shall be placed into a test chamber and subjected to helium gas at a minimum pressure of 30 pounds per square inch for a minimum period of 30 minutes. The battery shall then be placed into a container of water which is maintained at 160° ± 5°F. The battery shall remain in the water for a minimum period of 2 minutes, and it shall be visually inspected for leakage as evidenced by evolution of repeating bubbles. The battery shall comply with the requirements of 3.7 during this test.

4.4.5 Polarity. - The thermal battery shall be placed into a test chamber, the temperature of the chamber shall be raised to 160° ± 5°F., and it shall remain at this temperature for a minimum period of one hour (during the high-temperature test, the battery shall be tested after the 6-hour exposure). The polarity of the battery shall then be checked by use of a Government-approved electrometer, or other suitable high-impedance device, and it shall comply with the requirements of 3.8 during this test.

4.4.6 No-fire current. - Each thermal battery initiator shall be tested separately. A suitable voltmeter shall be connected across the output terminals of the battery (M-J to K-L) to monitor output voltage (to determine if the battery has fired). The test current shall be supplied from a suitable direct-current source. The battery shall comply with the requirements of 3.9 during this test.

4.4.7 Thermal battery operational tests. - During these tests, the thermal battery output voltage shall be monitored by means of a suitable recording oscillograph, recording voltage across pins MJ-KL, and the load circuit shall comply with the requirements of 3.10. Time duration shall be measured from the instant that activation current is applied.

4.4.7.1 Activation current. - The thermal battery activation current shall be supplied to initiator BG or CF by means of a suitable direct-current source. The battery shall comply with the requirements of 3.10.1 during this test.

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4.4.7.2 Activation time. - The activation time shall be measured by means of the recording chart specified in 4.4.7 and shall comply with the requirements of 3.10.2.

4.4.7.3 Load test. - The thermal battery shall be subjected to the loads specified in 3.10.3 after the battery is activated. The load voltages shall be measured by means of the recording chart specified in 4.4.7. The battery shall comply with the requirements of 3.10.3 during this test.

4.4.8 High temperature. - The thermal battery shall be placed into a test chamber, and the temperature of the chamber shall be raised to 160°F. The battery shall remain at this temperature for a minimum period of 6 hours and then tested as specified in 3.11 at the end of the high-temperature exposure period. The battery to be activated shall be allowed to cool to room temperature and then be subjected to the leakage test (see 3.7). After the leakage test, the battery shall be thoroughly dried and it shall be returned to the test chamber. The temperature of the chamber shall be raised to 160°F., and the battery shall remain at this temperature for a minimum period of 3 hours. At the end of this exposure period, the battery shall be subjected to the operational test (see 3.10). The battery shall comply with the requirements of 3.11 during this test.

4.4.9 Low temperature. - The thermal battery shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -65°F. The battery shall remain at this temperature for a minimum period of 6 hours and then tested as specified in 3.12 at the end of the low-temperature exposure period. The battery to be activated shall be allowed to warm to room temperature and then be subjected to the polarity test (see 3.8) and the leakage test (see 3.7). After the leakage test, the battery shall be thoroughly dried and it shall be returned to the test chamber. The temperature of the chamber shall be lowered to -65°F., and the battery shall remain at this temperature for a minimum period of 3 hours. At the end of this exposure period, the battery shall be subjected to the operational test (see 3.10). The battery shall comply with the requirements of 3.12 during this test.

4.4.10 Temperature shock. - The thermal battery shall be placed into a test chamber having an internal temperature of 160°F., and it shall remain at this temperature for a minimum period of 4 hours. At the end of this test exposure, and within a 5-minute period, the battery shall be transferred to a test chamber having an internal temperature of -65°F., and it shall remain at this temperature for a minimum period

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of 4 hours. This high- and low-temperature exposure constitutes one cycle, and the battery shall be subjected to three such cycles. The duration of exposure at each extreme temperature shall be not less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the completion of the third cycle (at the end of the low-temperature exposure period) the battery shall be removed from the test chamber and allowed to stabilize at room temperature. The battery shall then comply with the requirements of 3.13.

4.4.11 Vibration. - The thermal battery shall be securely mounted to a vibration machine using the four mounting brackets and mounting hardware actually used to mount the battery in the missile. The battery shall be subjected to the vibrations specified in Table V along the longitudinal axis and along one transverse axis. The battery shall comply with the requirements of 3.14 during this test.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation, packaging, and packing. - Preservation, packaging, and packing shall be level A or C as specified (see 6.1).

5.1.1.1 Level A and level C. - The thermal battery shall be preserved, packaged, and packed in accordance with instructions contained in MIL-STD-1169.

5.1.2 Marking. - Marking shall be in accordance with MIL-STD-1169.

5.2 Spare parts.

5.2.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. - The thermal battery shall be preserved and packaged in accordance with dwg. 8856849.

5.2.1.2 Level C. - The thermal battery shall be preserved and packaged in accordance with dwg. 8856849 except the barrier material may be omitted.

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5.2.2 Packing.

5.2.2.1 Level A and level C. - The thermal battery shall be packed in accordance with dwg. 8808362.

5.2.3 Marking. - Marking shall be as specified in accordance with dwg. 8808362 and 8856849.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

6.1.1 Procurement requirements:
(a) Title, number, and date of this specification.
(b) Production item or spare part procurement.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.1.2 Contract data requirements. - Data specified in 3.4, 3.5.1, 3.10.3 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664, or this specification, when specified on a DD Form 1423 incorporated into the contract.

6.2 Objective evidence. - Records of the contractor's quality control and inspections which can be verified.

6.3 Inspection of connectors.

6.3.1 Height of pins. - In the event a pin does not appear to be the same relative height of adjacent pins, a gage shall be used to determine the variation from normal height. Reject any connector that has a pin that varies by more than 0.040 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

6.4 Acceptance inspection equipment. - The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SMUPA-NR4.

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6.5 Safety precautions. - While performing electrical tests, the thermal battery shall be placed into an approved safety barrier to prevent possible injury to personnel or damage to equipment. After the battery is activated, the external case temperature reaches dangerous proportions and approved insulated gloves shall be used while handling the battery.

6.6 Supersession data. - This specification includes the requirements of Purchase Description ABMA-PD-B-973 (Rev. 2) dated 29 July 1963.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No.: 6135-A120

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